

THESIS
THESIS IMPLEMENTATION USING LATEX FOR
EFFECTIVENESS AT THESIS EXAM



Written by:
Muhammad Dafi Al Haq
NIM: 452024611067

DEPARTMENT OF INFORMATICS ENGINEERING
FACULTY OF SCIENCE AND TECHNOLOGY
UNIVERSITY OF DARUSSALAM GONTOR

2025/1447

ABSTRACT

In the midst of rapid developments in the digital age, access to scientific exegesis literature on the Qur'an is still limited, especially in an easily accessible and systematically organised digital format. However, scientific exegesis plays an important role in linking the verses of the Qur'an with scientific phenomena, thereby strengthening the integration between science and religious teachings. This research aims to develop a scientific exegesis website for the Qur'an that presents a collection of science-themed verses along with their exegesis, translations, and supporting references in a systematic manner. The development was carried out using the Model-View-Controller (MVC) architecture based on the Laravel framework, chosen for its ability to separate logic, presentation, and data, thereby facilitating system management and maintenance. The development method follows the water-fall model through the stages of requirements analysis, design, implementation, and testing. Testing includes expert validation by media experts, subject matter experts, user testing, and black-box testing to ensure all functions operate according to specifications. The test results showed a high level of feasibility, with scores of 86.20% from media experts, 87% from material experts, and 86.14% from users, while black-box testing proved that all features functioned properly. With these achievements, the website is considered feasible for use as a digital learning medium that supports the integration of science and religion. Recommendations for further development could focus on adding semantic search features and integrating interactive multi-media content to enhance the user experience.

Keywords: *science exegesis, integration of religion and science, laravel, model-view-controller (mvc), digitalization.*

AUTHORIZATION

THESIS IMPLEMENTATION USING LATEX IN TEXSTUDIO AT
DARUSSALAM GONTOR UNIVERSITY SIMAN

Case Study: Thesis Implementation

THESIS

Submitted to fulfill part of the requirement for obtaining a Bachelor of Computer
Science degree

Compiled by:

Muhammad Dafi Al Haq

NIM: 452024611067

This thesis has been tested and passed on
23 September 2025

Has been checked and approved by:

Supervisor 1,

Supervisor 2,

1st Lecture Name

NIY: 121212

2nd Lecture Name

NIY: 212121

Examiner,

Examiner Name

NIY: 111111

Known,

Head of Informatics Engineering,

Dean Name

NIY: 111111

STATEMENT OF ORIGINALITY

I declare truly that to the best of my knowledge, in this thesis manuscript there is no scientific work that has ever been submitted by another person to obtain an academic degree at a university, and there is no work or opinion that has ever been written or published by another person, except for those which are cited in writing in this manuscript and mentioned in the bibliography.

If it is proven that this thesis manuscript contains elements of plagiarism, I am willing for this thesis to be disqualified and the academic degree I have obtained (bachelor's degree) to be cancelled, and to be processed in accordance with applicable laws and regulations (Law No. 20 of 2003, Article 25 paragraph 2 and Article 70).

Ponorogo, 8 Agustus 2027

Muhammad Dafi Al Haq

NIM: 452024611067

ACKNOWLEDGEMENTS

Praise and gratitude to the presence of Allah SWT who has given His blessings and guidance so that the thesis entitled "THESIS IMPLEMENTATION USING LATEX" can be completed. This thesis is one of the requirements to complete the study and obtain a Bachelor of Computer Science degree in the Informatics Engineering Study Program, Darussalam Gontor University. The series of works on this thesis would not have been able to be completed until now without the help of various parties. Therefore, the author would like to express his gratitude for the help and support that has been given to:

1. Al-Ustadz Prof. Dr. Hamid Fahmy Zarkasyi, M.A.Ed., M.Phil., as the Chancellor of Darussalam Gontor University and all his staff, as well as all the lecturers of Darussalam Gontor University Ponorogo who have educated the author.
2. Al-Ustadz Dr. Haris Setyaningrum, M.Sc. as Dean of the Faculty of Science and Technology and his staff.
3. Al-Ustadz Dihin Muriyatmoko, M.T. as Head of the Informatics Engineering Study Program, Darussalam Gontor University.

The author realizes that this thesis is far from perfect due to limited experience. Therefore, the author welcomes constructive criticism and suggestions to improve this thesis. Finally, the author hopes that this paper can serve as a reference and resource for those who need it.

Ponorogo, 8 Agustus 2027

Muhammad Dafi Al Haq

NIM: 452024611067

TABLE OF CONTENTS

LIST OF TABLES

LIST OF FIGURES

CHAPTER I

INTRODUCTION

1.1 Background of the Study

The Qur'an is often considered a book that contains a lot of knowledge, both in the spiritual and scientific fields. One aspect of its miracle is the ability of the Qur'an to provide guidance regarding scientific phenomena that are relevant throughout time.¹

in Q.S. Fussilat (41:53), Allah said:

سَنُرِيهِمْ آيَاتِنَا فِي الْآفَاقِ وَفِي أَنْفُسِهِمْ حَتَّىٰ يَتَبَيَّنَ لَهُمْ أَنَّهُ الْحَقُّ ۗ أَوَلَمْ يَكْفِ بِرَبِّكَ أَنَّهُ عَلَىٰ كُلِّ شَيْءٍ شَهِيدٌ

Meaning: "We will show them Our signs in the horizons and within themselves until it becomes clear to them that it (the Qur'an) is the truth. Is it not enough (for you) that your Lord is a witness over all things?" (Q.S. Fuṣṣilat 41:53)

1.2 Problem Statement

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et,

¹bakir2016.

tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa.

Quisque ullamcorper placerat ipsum. Cras nibh. Morbi vel justo vitae lacus tincidunt ultrices. Lorem ipsum dolor sit amet, consectetur adipiscing elit. In hac habitasse platea dictumst. Integer tempus convallis augue. Etiam facilisis. Nunc elementum fermentum wisi. Aenean placerat. Ut imperdiet, enim sed gravida sollicitudin, felis odio placerat quam, ac pulvinar elit purus eget enim. Nunc vitae tortor. Proin tempus nibh sit amet nisl. Vivamus quis tortor vitae risus porta vehicula.

Fusce mauris. Vestibulum luctus nibh at lectus. Sed bibendum, nulla a faucibus semper, leo velit ultricies tellus, ac venenatis arcu wisi vel nisl. Vestibulum diam. Aliquam pellentesque, augue quis sagittis posuere, turpis lacus congue quam, in hendrerit risus eros eget felis. Maecenas eget erat in sapien mattis porttitor. Vestibulum porttitor. Nulla facilisi. Sed a turpis eu lacus commodo facilisis. Morbi fringilla, wisi in dignissim interdum, justo lectus sagittis dui, et vehicula libero dui cursus dui. Mauris tempor ligula sed lacus. Duis cursus enim ut augue. Cras ac magna. Cras nulla. Nulla egestas. Curabitur a leo. Quisque egestas wisi eget nunc. Nam feugiat lacus vel est. Curabitur consectetur.

Suspendisse vel felis. Ut lorem lorem, interdum eu, tincidunt sit amet, laoreet vitae, arcu. Aenean faucibus pede eu ante. Praesent enim elit, rutrum at, molestie non, nonummy vel, nisl. Ut lectus eros, malesuada sit amet, fermentum eu, sodales cursus, magna. Donec eu purus. Quisque vehicula, urna sed ultricies auctor, pede lorem egestas dui, et convallis elit erat sed nulla. Donec

luctus. Curabitur et nunc. Aliquam dolor odio, commodo pretium, ultricies non, pharetra in, velit. Integer arcu est, nonummy in, fermentum faucibus, egestas vel, odio.

Sed commodo posuere pede. Mauris ut est. Ut quis purus. Sed ac odio. Sed vehicula hendrerit sem. Duis non odio. Morbi ut dui. Sed accumsan risus eget odio. In hac habitasse platea dictumst. Pellentesque non elit. Fusce sed justo eu urna porta tincidunt. Mauris felis odio, sollicitudin sed, volutpat a, ornare ac, erat. Morbi quis dolor. Donec pellentesque, erat ac sagittis semper, nunc dui lobortis purus, quis congue purus metus ultricies tellus. Proin et quam. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Praesent sapien turpis, fermentum vel, eleifend faucibus, vehicula eu, lacus.

1.3 Objectives of the Study

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo.

Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa.

Quisque ullamcorper placerat ipsum. Cras nibh. Morbi vel justo vitae lacus tincidunt ultrices. Lorem ipsum dolor sit amet, consectetur adipiscing elit. In hac habitasse platea dictumst. Integer tempus convallis augue. Etiam facilisis. Nunc elementum fermentum wisi. Aenean placerat. Ut imperdiet, enim sed gravida sollicitudin, felis odio placerat quam, ac pulvinar elit purus eget enim. Nunc vitae tortor. Proin tempus nibh sit amet nisl. Vivamus quis tortor vitae risus porta vehicula.

Fusce mauris. Vestibulum luctus nibh at lectus. Sed bibendum, nulla a faucibus semper, leo velit ultricies tellus, ac venenatis arcu wisi vel nisl. Vestibulum diam. Aliquam pellentesque, augue quis sagittis posuere, turpis lacus congue quam, in hendrerit risus eros eget felis. Maecenas eget erat in sapien mattis porttitor. Vestibulum porttitor. Nulla facilisi. Sed a turpis eu lacus commodo facilisis. Morbi fringilla, wisi in dignissim interdum, justo lectus sagittis dui, et vehicula libero dui cursus dui. Mauris tempor ligula sed lacus. Duis cursus enim ut augue. Cras ac magna. Cras nulla. Nulla egestas. Curabitur a leo. Quisque egestas wisi eget nunc. Nam feugiat lacus vel est. Curabitur consectetur.

Suspendisse vel felis. Ut lorem lorem, interdum eu, tincidunt sit amet, laoreet vitae, arcu. Aenean faucibus pede eu ante. Praesent enim elit, rutrum at, molestie non, nonummy vel, nisl. Ut lectus eros, malesuada sit amet, fermentum eu, sodales cursus, magna. Donec eu purus. Quisque vehicula, urna sed ultricies auctor, pede lorem egestas dui, et convallis elit erat sed nulla. Donec luctus. Curabitur et nunc. Aliquam dolor odio, commodo pretium, ultricies non, pharetra in, velit. Integer arcu est, nonummy in, fermentum faucibus, egestas vel, odio.

Sed commodo posuere pede. Mauris ut est. Ut quis purus. Sed ac odio. Sed vehicula hendrerit sem. Duis non odio. Morbi ut dui. Sed accumsan risus eget odio. In hac habitasse platea dictumst. Pellentesque non elit. Fusce sed justo eu urna porta tincidunt. Mauris felis odio, sollicitudin sed, volutpat a, ornare ac, erat. Morbi quis dolor. Donec pellentesque, erat ac sagittis semper, nunc dui lobortis purus, quis congue purus metus ultricies tellus. Proin et quam.

Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Praesent sapien turpis, fermentum vel, eleifend faucibus, vehicula eu, lacus.

1.4 Significance of the Study

1. Significance for Practitioners

- (a) Significance 1
- (b) Significance 2
- (c) Significance 3

2. Significance for Researcher

- (a) Significance 1
- (b) Significance 2
- (c) Significance 3

3. Significance for University

- (a) Significance 1
- (b) Significance 2
- (c) Significance 3

1.5 Scope of Study

- 1. scope 1
- 2. scope 2
- 3. scope 3

1.6 systematic discussion

The systematics of writing in this research consists of five parts as follows:

CHAPTER 1: INTRODUCTION

1.1 Background of the Study

1.2 Problem Statement

- 1.3 Objectives of the Study
- 1.4 Significance of the Study
- 1.5 Scope of Study
- 1.6 systematic discussion

CHAPTER 2: LITERATURE REVIEW

- 2.1 Previous Studies
- 2.2 Theoretical Foundation

CHAPTER 3: RESEARCH METHODOLOGY

- 3.1 title
- 3.2 title
- 3.3 title
- 3.4 title

CHAPTER 4: RESULTS AND DISCUSSION

- 4.1 Result
- 4.2 Application Testing Results
- 4.3 Discussion

CHAPTER 5: CONCLUSION

- 5.1 Conclusion
- 5.2 Suggestions

CHAPTER II

LITERATURE REVIEW

2.1 Theoretical Foundation

2.1.1 teory 1

Modern software development is fundamentally guided by the principles of software engineering, which provide a systematic and disciplined approach to designing, developing, testing, and maintaining software systems. This structured methodology is essential for managing the complexity of large-scale projects, ensuring that the final product is reliable, efficient, and meets all specified requirements¹. Without adhering to these engineering principles, a project risks facing significant issues in scalability, maintainability, and overall quality, making it a critical theoretical cornerstone for any application development.

2.1.2 teory 2

Among the various architectural patterns available, the Model-View-Controller (MVC) stands out as a prevalent paradigm for organizing application logic, particularly in web development. The core strength of MVC lies in its separation of concerns, which divides the application into three interconnected components: the Model for data handling, the View for user interface presentation, and the Controller for managing user input and business logic². This separation not only enhances code organization but also facilitates parallel development and simplifies future modifications, a concept effectively demonstrated in early implementations of content management systems³.

2.1.3 teory 3

Beyond the internal architecture, the success of a software application is heavily dependent on its external qualities, primarily its functionality and usability. Functionality refers to the set of features and capabilities that the system provides to its users, while usability measures how easily and

¹sommerville2016.

²majeed2018.

³kristoko2009.

effectively users can interact with those features to achieve their goals⁴. A system that is functionally powerful but difficult to use will likely face poor user adoption. Therefore, a user-centered design approach that prioritizes both of these aspects is crucial for creating applications that are not only powerful but also intuitive and satisfying to use.

2.1.4 teory 4

The discourse on the relationship between science and religion provides a vital theoretical framework for this research. The concept of integrating and interconnecting religious sciences with empirical sciences aims to bridge the perceived gap between revelation and reason, proposing a holistic approach to knowledge⁵. This perspective challenges the notion that culture and religion are merely peripheral influences on science, arguing instead that they can be integral components in the pursuit of knowledge, shaping both the questions asked and the interpretations drawn⁶.

2.1.5 teory 5

The contemporary digital era has ushered in a profound transformation in how knowledge is disseminated and consumed, a phenomenon described as the "onlinization" of scholarly discourse⁷. This theoretical context is particularly relevant to the study of religious texts, where the migration from printed books to digital platforms has influenced both the development and the public's engagement with contemporary interpretations⁸. Understanding this digital shift is essential for analyzing the impact of new media on traditional fields of study and the new opportunities it creates for scholarly communication.

2.2 Previous Studies

2.2.1 Studies 1

Previous research in the Indonesian context has extensively documented the practical application of the MVC architecture across various domains. Studies have shown its successful implementation in building

⁴goodwin1987.

⁵muqoyyidin2014.

⁶muslih2010.

⁷maulana2021.

⁸sihabussalam2024.

adaptive online quiz systems for educational purposes, highlighting its flexibility⁹. Furthermore, its use in developing institutional websites, such as for vocational high schools, demonstrates its reliability for managing academic information systems¹⁰. The adoption of frameworks like CodeIgniter for e-commerce platforms further solidifies the pattern's standing as a go-to solution for structured web development in the region¹¹.

2.2.2 Studies 2

The development of e-commerce websites has been a popular subject of research, often utilizing modern PHP frameworks that inherently implement the MVC pattern. Multiple studies have showcased the use of Laravel for building online stores, from hijab shops to shoe stores, praising its robust features for creating complex and secure applications¹². These case studies often involve integrating third-party APIs, such as RajaOngkir for shipping cost calculation, to enhance the platform's functionality and provide a comprehensive user experience¹³.

2.2.3 Studies 3

The field of scientific Quranic exegesis in Indonesia has been explored from various epistemological standpoints. Research by Bakir (2016) analyzed the paradigm of contextualizing Quranic interpretation based on scientific findings, framing it within modern epistemology¹⁴. This approach aligns with broader scholarly efforts to interpret the Qur'an through the lens of modern science, a trend that has gained traction over the years¹⁵. The integration of the Qur'an and science has become a distinct epistemological approach within Indonesian Islamic scholarship, aiming to affirm the scripture's relevance in the modern world¹⁶.

⁹hidayat2012.

¹⁰wijaya2019.

¹¹suharsana2016.

¹²aipina2022; nugroho2021.

¹³prawito2020.

¹⁴bakir2016.

¹⁵laila2014.

¹⁶supriadi2018.

2.2.4 Studies 4

The rise of digital platforms has created a new frontier for Quranic studies, a development captured by recent bibliometric research. These studies map the growth of Al-Qur'an interpretation research in the digital era, identifying key trends and contributors using computational methods¹⁷. The digital transformation is also evident in comparative studies of online tafsir platforms, which analyze how different websites present and structure Quranic knowledge for a digital audience¹⁸. This body of research underscores the irreversible shift towards digital mediums for religious scholarship.

2.2.5 Studies 5

On an international level, the scientific approach in Quranic exegesis has been a subject of continuous academic discussion, exploring its emergence and related issues¹⁹. Scholars have examined the contributions of various individuals, institutions, and even governments in promoting scientific readings of the Qur'an²⁰. This integration of scientific knowledge and religious sciences is not merely a theoretical exercise but also has practical implications for Islamic education, aiming to foster a holistic understanding among students²¹.

¹⁷**fadhilah2021.**

¹⁸**azizah2024.**

¹⁹**ismail2021.**

²⁰**asnawi2021.**

²¹**nasir2022.**

CHAPTER III

RESEARCH METHODOLOGY

3.1 Time and Location of research

Day & Date : Monday, 17 June 2027 Place : ponoroggo siman

3.2 Research Tools and Materials

The tools and materials we used in this research are:

- **Tools:**

1. **tool 1**, Detail.
2. **tool 2**, Detail.
3. **tool 3**, Detail.

- **Materials:**

1. **material 1**, Detail.
2. **material 2**, Detail.
3. **material 3**, Detail.

3.3 Research Model

The research model employed for the development of this web-based platform follows a structured design methodology, encompassing several distinct phases from requirements analysis to final implementation. The initial phase involved a detailed analysis of system functionalities, which were defined through the creation of use case diagrams and their corresponding scenarios to capture user interactions accurately¹. Following this, the system architecture was designed using the Model-View-Controller (MVC) pattern to ensure a clear separation between the data layer, business logic, and the user interface, a common practice in building adaptive web systems². The implementation phase was then carried out using the Laravel framework, a modern toolset well-suited for developing online information systems³. This phased approach is aligned

¹taufan2022.

²hidayat2012.

³prawito2020.

with established software engineering principles, providing a systematic workflow from concept to a functional application⁴. all of that is in accordance with figure ??



Figure III.1: ini cuma contoh

3.4 Research Stages

3.4.1 Requirement Analysis

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

3.4.2 Design

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum

⁴sommerville2016.

gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

3.4.3 Implementation

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

3.4.4 Verification (Testing)

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius

orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

3.4.5 Maintenance

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

CHAPTER IV

RESULTS AND DISCUSSION

4.1 Result

4.1.1 Description 1

The evolution of modern educational materials has seen a significant shift towards interactive technologies, with methods such as QR code integration being developed to enhance discovery learning in topics like mathematical probability¹. However, the success of such technological integration is not merely dependent on its novelty, but on the core principles of functionality and usability, which dictate how effectively a user can interact with the system to achieve a desired outcome². Underpinning these user-facing features is the critical importance of a well-designed data structure, often a relational database, which is fundamental for managing the resources, whether for an online bookstore or a complex learning platform³. or if summarized as in table ?? below.

Table IV.1: contoh tabel.

Framework	Bahasa	Pola Arsitektur	Tahun Rilis
Laravel	PHP	MVC	2011
Django	Python	MVT (Mirip MVC)	2005

4.1.2 Description 2

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel

¹ramadhan2024.

²goodwin1987.

³alhusain2016.

leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

4.1.3 Description 3

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

4.2 Application Testing Results

4.2.1 Internal Testing

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

4.2.2 External Testing

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum

sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

4.3 Discussion

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

CHAPTER V

CONCLUSION

5.1 Conclusion

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

5.2 Suggestions

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.